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Academic Year: 2025-2026 Even

Department of Artificial Intelligence and Machine Learning

Date: 15/04/2026	Test – 1	Max. Marks : 10 + 50
Semester: VI	UG	Duration : 120 mins
Course Title: Generative Artificial Intelligence		Course Code: AI365TDD

Part – A

Q. No	Question	M	BT	CO
1.1	In a Variational Autoencoder (VAE), why do we learn a distribution (mean and variance) instead of a single latent vector?	2	2	1
1.2	Define the probability functions used in discriminative and generative models.	2	1	1
1.3	List any two limitations of Large Language models (LLMs).	2	1	3
1.4	What is the role of the embedding layer in Large Language Models?	2	2	1
1.5	In Variational Autoencoders (VAEs), why is the loss function defined as a combination of two terms instead of a single objective?	2	2	1

Part – B

Q. No	Question	M	BT	CO
1	Construct a Transformer-based encoder–decoder model that uses: • Self-attention • Masking • Positional encoding Draw a neat architecture diagram and explain each component involved in encoding and decoding the input sequence.	10	4	2
2	An autoencoder is used to compress image data through its encoder. Consider an input image of size **28 × 28 × 1**. The encoder consists of convolutional layers with the following parameters: Layer 1: 16 filters, kernel = 3×3, stride = 2, padding = 1 Layer 2: 32 filters, kernel = 3×3, stride = 2, padding = 1 Write a Python program (using PyTorch or TensorFlow) to implement the encoder.	10	3	2

3a	How does representation learning simplify operations on data (such as modifying attributes) compared to directly manipulating input space? Explain with a suitable example.	5	4	3
3b	Implement latent space sampling for a Variational Autoencoder using the reparameterization trick, without directly sampling from a normal distribution at the output layer. Your answer should include: • Intermediate transformations steps • Architecture block diagram • Graphical illustration of sampling	5	3	2
4a	Autoencoders are often used as denoising models by learning compact latent representations. (i) Explain how an autoencoder removes noise during training. (ii) Analyze the effect of the following on reconstruction quality: • A very small latent space • A very large latent space	5	3	3
4b	An encoder–decoder model is trained to reconstruct input data. Another variant of the model introduces a constraint on the latent space distribution. (i) How would you evaluate the performance of both models during training? (ii) Explain the components of their loss functions used in each case. (iii) Write a Python snippet to compute the reconstruction loss for the first model.	5	3	2
5	An encoder–decoder model maps each input to a fixed latent vector and achieves high reconstruction accuracy but poor generalization. (i). Identify this class of models. (ii). Explain how introducing a distribution over latent variables modifies the objective function. Provide the mathematical formulation. (iii). List two differences between this model and its probabilistic variant (Variational Autoencoder).	10	4	4

Department of Artificial Intelligence and Machine Learning

Date: 23/5/2026	**Test – II**	**Max. Marks: 10 + 50**
Semester: VI	**UG**	**Duration: 120 mins**
Course Title: Generative Artificial Intelligence		**Course Code: AI365TDD**

Part – A

Q. No	Question	M	BT	CO
1.1	List any two normalization techniques used in deep learning. Also identify the normalization technique commonly used in CycleGAN architectures?	2	1	1
1.2	Why is the parameter β_t important in the forward diffusion process? Write the complete equation of the forward diffusion process	2	2	1
1.3	What is meant by "cycle consistency" in CycleGAN?	2	2	3
1.4	State the mathematical expression for the Style Loss used in Neural Style Transfer.	2	2	1
1.5	During neural style transfer, why is total variation loss added even after optimizing the content and style losses?	2	2	1

Part – B

Q. No	Question	M	BT	CO
1.a	Explain how PatchGAN discriminators differ from traditional GAN discriminators and why they are effective for image translation tasks	5	2	2
1.b	Compare linear diffusion schedules and cosine diffusion schedules with respect to noise addition and image quality.	5	3	2
2.a	Describe the reparameterization trick in diffusion models and explain why it significantly improves computational efficiency.	5	3	3
2.b	Deep image translation networks require specialized generator architectures to preserve image details while learning complex transformations. Identify a suitable generator architecture used in CycleGAN for this purpose and explain its architecture with a neat block diagram. Also discuss why such an architectures are required in deep networks and explain the major training problem it helps to overcome.	5	3	2
3	Explain the working of U-Net architecture used in a CycleGAN. Discuss the role of concatenation and skip connections in the architecture. Also explain the major challenges faced during GAN training and analyze their impact on model performance.	10	3	3
4	Discuss the forward diffusion process and reverse diffusion process in detail.	10	3	2
4	Write a Python program using Keras to design and compile the complete combined CycleGAN model. The implementation should include appropriate handling of generator training, discriminator freezing, reconstruction learning, identity preservation, and validity prediction for both domains.	10	4	4

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Date: 18.06.2026	**Improvement Test**	**Max. Marks: 50+10**
Semester: VI	**UG**	**Duration: 1 $\frac{1}{2}$ Hrs + ½ Hr**
Course Title: Generative Artificial Intelligence		**Course Code: AI365TDD**

Part A

Q. No		PART A Improvement QUIZ Questions	M	BT	CO
1	a)	Define Energy-Based Model (EBM)?	2	1	3
	b)	How does Langevin Dynamics help in sampling for Energy-Based Models	2	2	3
	c)	Define **algorithmic bias** and **data bias** in the context of generative AI	2	1	4
	d)	What is **statistical parity**, and how is it used to measure fairness in AI models?	2	2	5
	e)	List any two techniques used to reduce bias in AI models.	2	2	5

Part B

Q. No	PART B Improvement Test Questions	M	B	CO
1	How does Langevin Sampling enable image generation in Energy-Based Models? Consider an EBM trained on handwritten digits and illustrate the process mathematically.	10	2	3
2	Given an Energy-Based Model represented by the energy function E(x,θ): a. Explain how the model learns the underlying data distribution through energy minimization. (5M) b. Compare EBMs with likelihood-based generative models regarding probability estimation and sample generation. (5M)	10	3	3
3	A company deploys a Generative AI model for automated content generation. a. Examine how historical and societal biases can become embedded in AI-generated outputs. (5M) b. Illustrate with real-world examples where AI bias resulted in ethical or legal challenges. (5M) c. Recommend strategies for continuous monitoring and bias reduction after deployment. (5M)	15	4	4
4	Fairness in AI is measured using different metrics such as **statistical parity, equal opportunity, and disparate impact**. a. Compare these fairness metrics in terms of their applicability and limitations in generative AI models. (5m) b. Evaluate the effectiveness of **pre-processing, in-processing, and post-processing** mitigation strategies in addressing bias. (5m) c. Suggest an optimal bias mitigation strategy for a generative AI model used in hiring decisions and justify your choice. (5m)	15	4	5